

Prior-Data Fitted Networks

[s26] Theoretical Foundations of Amortization and Meta-, and In-Context Learning

Egor Kashkarov

July 4, 2026

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- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

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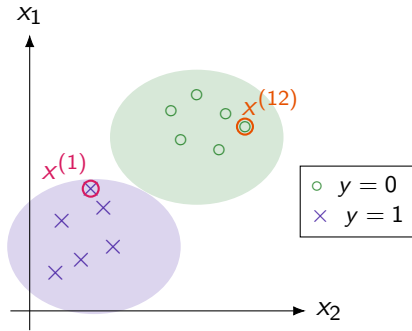
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Background

Statistical Model



$X \in \mathcal{X}$ – features; $Y \in \mathcal{Y}$ – classification label

r.v. $\mathcal{D}_n = (X^{(i)}, Y^{(i)})_{i=1}^n$ is s.t. $(X^{(i)}, Y^{(i)}) \stackrel{\text{iid}}{\sim} p_0(\cdot, \cdot)$

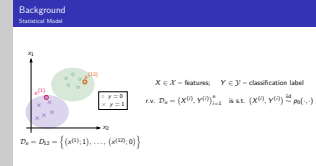
$$\mathcal{D}_n = \mathcal{D}_{12} = \left\{ (x^{(1)}; 1), \dots, (x^{(12)}; 0) \right\}$$

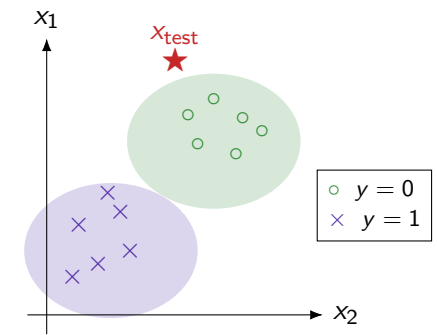
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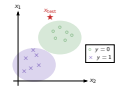




- At inference we want to estimate $p_0(Y | X)$:
- 1 fix $\mathcal{D}_n = D_n$
 - 2 train $q_\theta(X) \approx p_0(Y | X)$ on data from D_n
 - 3 predict $\mathbb{P}(Y = \hat{y}_{\text{test}} | X = x_{\text{test}}) = q_\theta(\hat{y}_{\text{test}} | x_{\text{test}})$

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Background
Supervised Learning Framework



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Background

Bayesian Framework

For every n , the statistical model gives a tuple $(\mathcal{D}_n \cup (Y, X), p)$.

We want to approximate $p(Y \mid X)$ by

$$\pi(y \mid x, D_n) \stackrel{\text{def}}{=} \mathbb{P}(Y = y \mid X = x; \mathcal{D}_n = D_n)$$

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$$q_\theta(X, \mathcal{D}_n) \approx \pi(Y \mid X, \mathcal{D}_n)$$

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And make predictions

$$\mathbb{P}(Y = \hat{y}_{\text{test}} \mid X = x_{\text{test}}, \mathcal{D}_n = D_n) = q_\theta(\hat{y}_{\text{test}} \mid x_{\text{test}}, D_n)$$

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π is called the **Posterior Predictive Distribution**

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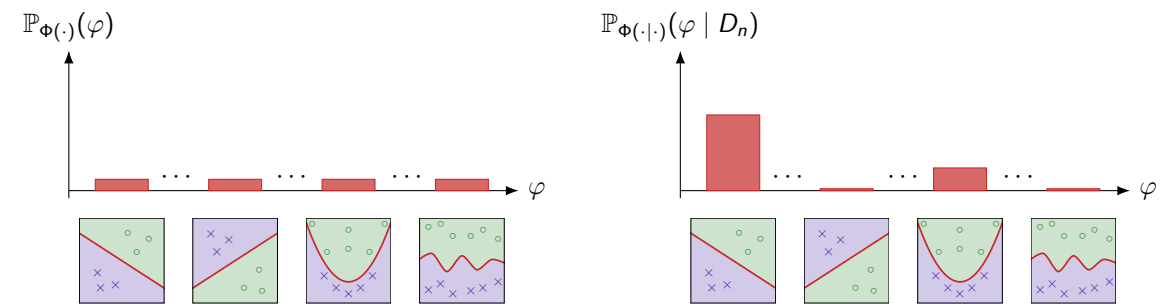
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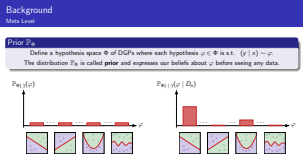
Meta Level

Prior $\mathbb{P}_\Phi$

Define a hypothesis space Φ of DGPs where each hypothesis $\varphi \in \Phi$ is s.t. $(y \mid x) \sim \varphi$.
The distribution $\mathbb{P}_\Phi$ is called **prior** and expresses our beliefs about φ before seeing any data.



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$$\pi(y \mid x, D_n) = \int \varphi(y \mid x) d\mathbb{P}_{\Phi(\cdot|\cdot)}(\varphi \mid x, D_n) \tag{1}$$

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$$\begin{aligned} &= \mathbb{P}(\varphi = \varphi_1 = \begin{array}{|c|} \hline \text{green} \\ \hline \text{blue} \end{array} \mid x, D_n) \cdot \varphi_1(y \mid x) \\ &+ \mathbb{P}(\varphi = \varphi_2 = \begin{array}{|c|} \hline \text{blue} \\ \hline \text{green} \end{array} \mid x, D_n) \cdot \varphi_2(y \mid x) + \dots \end{aligned}$$

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Posterior Predictive Distribution

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Assumption

$\mathbb{P}(\varphi \mid \mathbf{x}, D_n) = \mathbb{P}(\varphi \mid D_n)$ i.e. $\mathbf{x} \equiv \mathbf{x}_{\text{test}}$ is not informative about hypothesis φ .

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Dropping x from the conditioning on φ :

$$\pi(y \mid x, D_n) = \int \varphi(y \mid x) \, d\mathbb{P}_{\Phi(\cdot|\cdot)}(\varphi \mid D_n) = \int p(y \mid x, \varphi) \cdot \underbrace{p(\varphi \mid D_n)}_{\text{not available}} \, d\varphi$$

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Prior-Data Fitted Networks (PFNs)

Prior-Data Fitted Network is a numerical approximation of the entire *family* of PPDs $\{\pi(\cdot \mid \cdot, \mathcal{D}_n)\}_n$.

Theorem 2.2. [1]

Let $\mathcal{Q}$ be the set of *all* conditional probability functions. Then the PPD satisfies

$$\pi = \arg \max_{q \in \mathcal{Q}} \mathbb{E}_{\varphi \sim \mathbb{P}_{\Phi}, \mathcal{D}_n \cup (Y, X) \sim (\mathcal{D}_{n+1} | \varphi)} [\log q(Y \mid X, \mathcal{D}_n)].$$

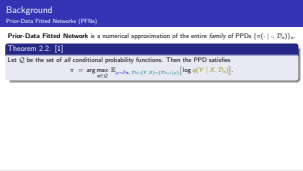
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Prior-Data Fitted Networks

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The expectation in Theorem 2.2 follows the hierarchical data-generating mechanism: first draw the true model (data-generating process) $\varphi \sim \Phi$; then generate the training sample $\mathcal{D}_n$ and the test pair (Y, X) from that selected φ .



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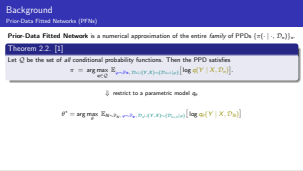
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$$\hat{\theta} = \arg \max_{\theta} \sum_{j=1}^m \log q_{\theta}(Y_j \mid X_j, D^{(j)})$$

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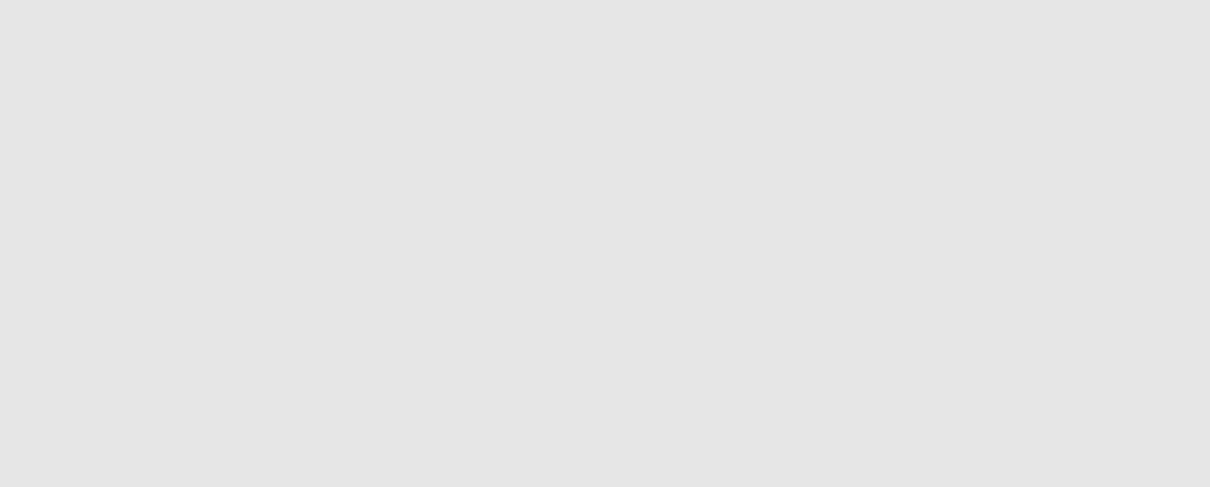
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Limitations

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Statistical Foundations of PFNs

Studied Questions

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When PPDs can Learn

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Statistical Foundations of PFNs

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PFN Approximation of the PPD

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Bias-Variance Decomposition of PFNs

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Variance and Diminishing Sensitivity

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Statistical Foundations of PFNs

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Localized PFNs

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Statistical Foundations of PFNs

Statistical Foundations of PFNs

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Background

Definition

Insert key definition here.

Key insight

Insert key insight here.

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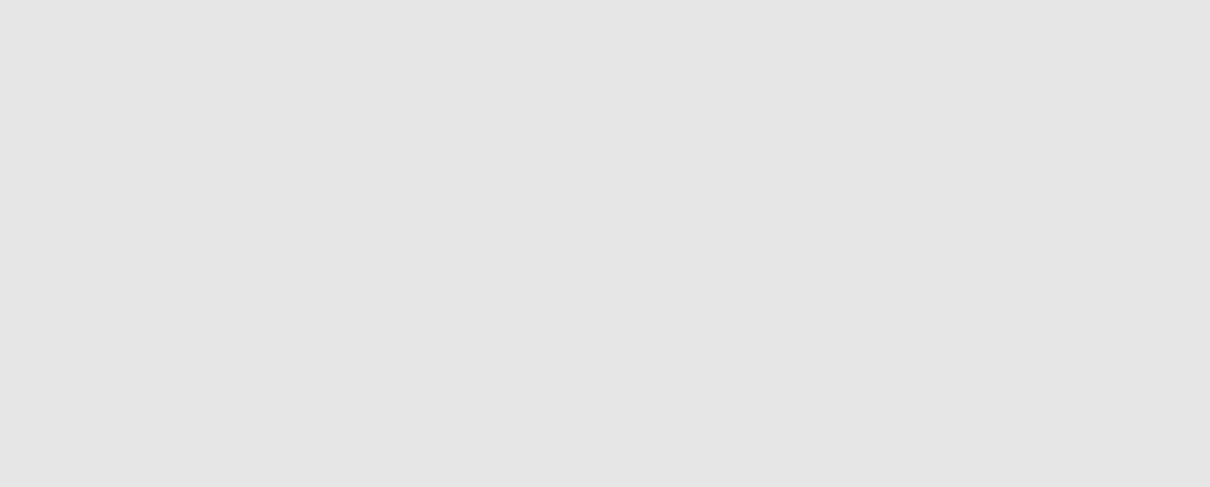
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Connections to Other Topics

Background: Simulation-Based Inference

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		Connections to Other Topics
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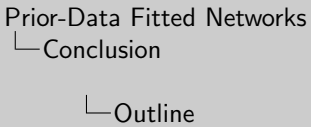
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 [Thomas Nagler.](#)
Statistical Foundations of Prior-Data Fitted Networks.
In Proceedings of the 40th International Conference on Machine Learning, pages
25660–25676. PMLR, July 2023.

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References

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25660–25676. PMLR, July 2023.